Lakshya JEE 2.0 2025

Maths

DPP: 4

Relations and Functions

Q1 For which of the following function, domain is not complete set of real numbers?
(A) $f(x)=x^2+x+1$
(B) $f(x)=\frac{1}{x^2-3x+4}$
(C) $f(x)=\frac{1}{x^2+1}$
(D) $f\left(x\right)=\frac{1}{x^2-5x+4}$

Q2 Range of the function $f\left(x\right)=\frac{x^2+x+2}{x^2+x+1}; x\in R$ is
(A) $(1,\infty)$
(B) $(1,11/7)$
(C) $(1,7/3]$
(D) $(1,7/5)$

Q3 Range of function $f(x)=\frac{x^2-3x+2}{x^2+x-6}$ is $R-\left\{\frac{a}{b},c\right\}$ What is $a+b+2c$?
(A) 4 (B) 5
(C) 6 (D) 8

Q4 If $f(x)=\frac{2x-3}{5-7x}$, then find domain and range of $f(x)$.

Q5 Find maximum and minimum value of $f(x)=12\cos x-5\sin x$

Q6 If x is real, then the expression $\frac{x+2}{(2x^2+3x+6)}$ takes all values in the interval
(A) $\left(\frac{1}{13},\frac{1}{3}\right)$
(B) $\left[-\frac{1}{13},\frac{1}{3}\right]$
(C) $\left(-\frac{1}{3},\frac{1}{13}\right)$
(D) $\left(-\infty,-\frac{1}{13}\right]\cup\left[\frac{1}{3},\infty\right)$

Q7 Find domain of $f(x)=\sqrt{1-\sqrt{1-x^2}}$.

Q8 Find Range of $f(x)=\sqrt{25-x^2}$

Q9 The domain of the function $f\left(x\right)=\sqrt{x^2-5x+6}+\sqrt{2x+8-x^2}$, is
(A) $[2,3]$
(B) $[-2,4]$
(C) $[-2,2]\cup[3,4]$
(D) $[-2,1]\cup[2,4]$

Q10 If $A=\{1,2,3\},\mathrm{B}=\{x,y\}$, then the number of functions that can be defined from A into B is
(A) 12 (B) 8
(C) 6 (D) 3

Q11 Let A be a set containing 10 distinct elements, then the total number of distinct functions from A to A is
(A) $10\,!$ (B) 10^{10}
(C) 2^{10} (D) $2^{10}-1$

Q12 Range of $\mathrm{f}(\mathrm{x})=\log_{\sqrt{5}}\left(\sqrt{2}(\sin\ \mathrm{x}\ -\cos\ \mathrm{x})+3\right)$ is
(A) [0, 1] (B) [0, 2]
(C) [0, 3/2] (D) [1, 2]

Q13 Let A be a finite set containing n distinct elements. The number of functions that can be defined from A to A is
(A) 2^n
(B) n^2
(C) $n\,!$
(D) n^n

Q14 Find the domain of the function
$f(x)=\sqrt{(9x-x^2)}$
(A) $(9,\infty)$
(B) $[0,9]$
(C) $(-\infty,0]$
(D) $[0,\infty)$

Q15 Find the domain of the function
$f\left(x\right)=\sqrt{\left(\frac{x-2}{x+2}\right)}+\sqrt{\left(\frac{1-x}{1+x}\right)}$.
(A) $(-\infty,-2]\cup[2,\infty)$
(B) $[-1,1]$
(C) ϕ
(D) None of these

Q16 Domain of $f(x)=\sqrt{\log_2\log_{1/3}\log_{1/4}x}$, is:
(A) $x\in\left[\left(\frac{1}{3}\right)^{1/4},1\right]$
(B) $x\in\left[\left(\frac{1}{4}\right)^{1/3},1\right)$
(C) $x\in\left[3^4,1\right]$
(D) $x\in\left[\left(\frac{1}{4}\right)^{3},1\right]$

Q17 Domain of $y=\log_{(x-4)}\left(x^2-11x+24\right)$ is:
(A) $x\in(5,\infty)$
(B) $x\in(7,\infty)$
(C) $x\in(4,\infty)$
(D) $x\in(8,\infty)$

Q18 Domain of : $f(x)=\sqrt{\log_{\frac{1}{2}}\left(\frac{x-1}{x+5}\right)}$

Q19 A function f from the set of natural numbers to integers defined by
$f(n)=\begin{cases}\frac{n-1}{2}, & \text{when } n \text{ is odd}\\ -\frac{n}{2}, & \text{when } n \text{ is even}\end{cases}$, is:
(A) One-one but not onto
(B) Onto but not one-one
(C) One-one and onto both
(D) Neither one-one nor onto

Q20 The function $f:R\to R$ defined by $f(x)=e^x$ is:
(A) Onto
(B) Many-one
(C) One-one and into
(D) Many one and onto

Q21 Function $f:N\to N, f(x)=2x+3$ is:
(A) One-one onto
(B) One-one into
(C) Many-one onto
(D) Many-one into

Q22 If $A=\{a,b,c\}$, then total number of one-one onto functions which can be defined from A to A is:
(A) 3
(B) 4
(C) 9
(D) 6

Q23 Find number of surjection from A to B where $A=\{1,2,3,4\},\ B=\{a,b\}$
(A) 13
(B) 14
(C) 15
(D) 16

Answer Key

Q1 (D)

Q2 (C)

Q3 (D)

Q4 Domain = R – $\left\{\frac{5}{7}\right\}$
Range = R – $\left\{-\frac{2}{7}\right\}$

Q5 maximum = 13, minimum = - 13

Q6 (B)

Q7 $x \in [-1, 1]$

Q8 $R_f \in [0, 5]$

Q9 (C)

Q10 (B)

Q11 (A)

Q12 (B)

Q13 (D)

Q14 (B)

Q15 (C)

Q16 (B)

Q17 (D)

Q18 $x \in (1, \infty)$

Q19 (C)

Q20 (C)

Q21 (B)

Q22 (D)

Q23 (B)

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